***Abstract*—**
***Index Terms*—**

I. INTRODUCTION

Code-mixed, Roman-script social media text—commonly termed *Banglish*, in which Bangla is transliterated into Latin characters and freely interleaved with English—is ubiquitous across South Asian social platforms, yet it is poorly served by standard NLP tooling trained on formal, monolingual corpora. The core obstacle is orthographic: Banglish has no fixed spelling convention, so a single word may surface in a dozen visually and phonetically related but string-dissimilar forms, undermining tokenizers, embeddings, and classifiers that assume orthographic regularity.

We focus this problem on sentiment classification under a low-label-budget regime—the realistic deployment scenario in which thousands of unlabeled comments are readily scraped but only a small fraction can be manually annotated. We instantiate this concretely with 600 labeled examples against a pool of roughly 9,430 available training rows, a labeling rate of approximately 6.4%, reflecting the annotation budgets typical of small research teams and low-resource languages rather than the fully-labeled corpora assumed by most benchmark work.

Prior Banglish sentiment classification work has largely pursued one of two directions: data augmentation to synthetically expand labeled sets [1, 2], or fine-tuning deep transformer models on whatever labeled data is available [3]. Neither line evaluates classical two-view co-training in the original sense of Blum and Mitchell [4]—training two conditionally-independent-view classifiers that iteratively pseudo-label each other's confident predictions—as a low-resource remedy for this specific language setting, and no prior study rigorously diagnoses why co-training helps when it does. In particular, co-training gains in the literature are typically presented as a single aggregate improvement, without decomposing how much of that gain comes from ensembling two views at all versus from the iterative pseudo-labeling process itself—a distinction this paper makes explicit.

This paper makes the following contributions:

- We reproduce and rigorously benchmark a two-view (orthographic + phonetic) co-training pipeline for Banglish sentiment classification, validating our harness against the original implementation to within 0.003 macro-F1.
- We decompose the co-training gain into an ensemble component (+0.015, from combining two views at iteration 0) and an iteration component (+0.01), showing the latter is smaller than commonly assumed.
- We show the deployed pseudo-label agreement gate's purity advantage is not stable across iterations—it wins at iteration 0 (0.929 vs. 0.916) and loses by the final iteration (0.854 vs. 0.891)—and attribute this to declining view independence as pseudo-labels accumulate in both pools.
- We conduct an exhaustive comparison of 21 classical classifiers across 7 ML families and find a previously untested classifier (`CalibratedClassifierCV(RidgeClassifier)`) beats the deployed choice on both views, with the advantage verified stable across label budgets (300–1,200) and under synthetic spelling noise (up to 35%).
- We identify and quantify a hard performance ceiling ($\sim$0.66 macro-F1 under full supervision) attributable to label-scheme noise, not model choice.

The remainder of the paper is organized as follows: Section II reviews related work; Section III describes the dataset; Section IV details the methodology; Section V reports results; Section VI discusses their implications; and Section VII concludes.

II. RELATED WORK

A. Co-training and semi-supervised learning theory

Co-training was introduced by Blum and Mitchell [4], who formalized the setting in which each labeled example can be described by two distinct feature views, and showed that if each view is individually sufficient to predict the label and the two views are conditionally independent given the label, a classifier trained on one view can generate confident pseudo-labels that usefully augment training data for a classifier trained on the other. This conditional-independence assumption is rarely satisfied exactly in practice, and Van Engelen and Hoos [5], in their survey of the broader semi-supervised learning landscape, identify the erosion of view independence over successive self-training or co-training rounds as a persistent, under-examined failure mode—one that most applied papers acknowledge in passing but few measure directly on a deployed pipeline. This is precisely the gap our paper closes: rather than reporting an aggregate co-training gain, we track the pseudo-label agreement gate's purity iteration-by-iteration and show its advantage over the alternative literally reverses sign as the two views' pseudo-label pools converge on each other's outputs—an empirical instance of the theoretical concern Van Engelen and Hoos raise, observed on real Banglish data rather than in simulation.

B. Banglish and code-mixed sentiment analysis

Recent work on Bengali–English code-mixed sentiment has pursued two main strategies for coping with label scarcity and orthographic noise. Tareq et al. [1] and Sultana and Mamun [2] both rely on data augmentation—generating synthetic training examples via cross-lingual word replacement and related transformations—to expand small labeled sets and to expose the model to spelling variation. Separately, Alam et al.

[3] introduce BnSentMix, a diverse 20,000-example Bengali–English code-mixed sentiment dataset, and benchmark 14 baselines including transformer encoders further pretrained on code-mixed text, reaching 69.8% accuracy (69.1% F1) under full supervision. We position our work against this backdrop rather than in competition with it: our classical two-view pipeline, trained on only 600 labeled examples with no GPU, is not intended to surpass transformer accuracy, but to approach it at a small fraction of the label budget and zero training cost—a materially different and, for low-resource deployment settings, arguably more actionable trade-off. Where Tareq et al. and Sultana and Mamun treat orthographic noise as something to be countered with synthetically augmented spelling variants, our weakly-supervised noise model instead learns and injects noise patterns from the pipeline's own disagreement signal, which we argue is a more targeted mechanism than augmentation applied independent of what the model actually confuses.

C. Feature engineering and classifier calibration

Our View A representation builds directly on Cavnar and Trenkle [6], who showed that character n-gram features are robust to the spelling and typographical variation that defeats word-level tokenization—the same property that motivates their use here for Banglish, where a single word can surface in many orthographic forms. Our View B phonetic encoding draws on the classical Soundex family of phonetic coding schemes [7, 8], adapted with Bangla-specific phoneme groupings. For the classifier comparison in Section V, we ground our calibration discussion in two foundational results: Platt [9], who introduced fitting a sigmoid to SVM decision scores via an internal cross-validation loop to produce probability estimates—the mechanism responsible for the substantial slowdown of `SVC(probability=True)` relative to margin-only variants—and Niculescu-Mizil and Caruana [10], who showed empirically that several standard classifiers, including SVMs and boosted trees, produce poorly calibrated probability estimates out of the box, such that an uncalibrated margin-to-probability mapping can be a materially unreliable confidence signal. These two results provide the literature basis for our finding that `CalibratedClassifierCV`-wrapped classifiers outperform their uncalibrated counterparts specifically on the confidence-threshold-gated pseudo-labeling step central to co-training, rather than leaving this as an isolated empirical observation.

Taken together, these three clusters motivate the structure of the paper: Cluster A supplies the theoretical prediction (declining view independence) that our diagnostic in Sections V and VI tests directly; Cluster B supplies the applied baseline and the augmentation-versus-learned-noise contrast that frames our contribution honestly rather than as an unqualified improvement; and Cluster C supplies the mechanistic explanation for why calibration, not classifier family alone, drives the exhaustive comparison's result.

TABLE I
MAIN RESULTS: MACRO-F1 ON THE HELD-OUT TEST SET, MEAN OVER 3 SEEDS (42, 7, 2024).

Model	Macro-F1
Single-view SVM (baseline)	0.5534
Standard co-training (no WNM)	0.5767
PMVC-WNM (deployed)	0.5794
Best exhaustive candidate, View A (`ridge_calibrated`)	0.5652
Best exhaustive candidate, View B (`ridge_calibrated`)	0.5542
Full-supervision ceiling ($\sim$9,430 labels)	0.6562

TABLE II
VIEW A / VIEW B CLASSIFIER COMPARISON (MEAN OVER 3 SEEDS). LOWER BRIER SCORE INDICATES BETTER-CALIBRATED CONFIDENCE ESTIMATES.

View	Classifier	Macro-F1	Brier	Fit (s)
A	`svm_platt`	0.5624	0.1807	1.435
A	`logreg`	0.5613	0.1802	0.729
A	`svm_calibrated`	0.5596	0.1813	0.120
A	`svm_softmax`	0.5534	0.1960	0.018
B	`logreg`	0.5522	0.1845	1.314
B	`svm_softmax`	0.5488	0.2002	0.007
B	`complement_nb`	0.5340	0.1952	0.005
B	`random_forest`	0.5245	0.1907	1.492

III. DATASET

IV. METHODOLOGY

V. RESULTS

VI. DISCUSSION

VII. CONCLUSION

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[5] J. E. Van Engelen and H. H. Hoos, "A survey on semi-supervised learning," *Machine Learning*, vol. 109, no. 2, pp. 373–440, 2020.

[6] W. B. Cavnar and J. M. Trenkle, "N-gram-based text categorization," in *Proceedings of SDAIR-94, 3rd Annual

TABLE III
FEATURE N-GRAM RANGE COMPARISON AT THE DEPLOYED 5,000-FEATURE BUDGET (MEAN OVER 3 SEEDS).

View	Config	Macro-F1
A	char(2,3)	0.5611
A	char(3,4) [deployed]	0.5534
B	word(1,1)	0.5575
B	word(1,2) [deployed]	0.5522

TABLE IV
PSEUDO-LABEL GATE COMPARISON, CONFIDENCE THRESHOLD FIXED AT 0.55 (MEAN $\pm$ STD OVER 3 SEEDS).

Gate	Macro-F1	Pseudo-labels	Purity
agreement_and_confident (pmvc_wnm)	0.5794 $\pm$ 0.0094	7191	0.859
view_a_only (standard)	0.5767 $\pm$ 0.0101	5783	0.881
agreement_or_one_very_confident	0.5739 $\pm$ 0.0058	21827	0.757
agreement_no_threshold	0.5701 $\pm$ 0.0077	35100	0.709
or_union	0.5701 $\pm$ 0.0077	35100	0.673

Symposium on Document Analysis and Information Retrieval, 1994, pp. 161–175.
[7] R. C. Russell, “Index,” U.S. Patent 1,261,167, Apr. 1918.
[8] ——, “Index,” U.S. Patent 1,435,663, Nov. 1922.
[9] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” *Advances in Large Margin Classifiers*, vol. 10, no. 3, pp. 61–74, 1999.
[10] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, 2005, pp. 625–632.

TABLE V
CUMULATIVE SINGLE-VIEW (VIEW A) IMPROVEMENT LADDER, EACH ROW ADDS ONE CHANGE ON TOP OF THE PREVIOUS (MEAN OVER 3 SEEDS).

Configuration	Macro-F1
0. baseline (svm_softmax, char(3,4), 5k feat)	0.5534 $\pm$ 0.0120
1. max_features=50000	0.5729 $\pm$ 0.0036
2. + min_df=2	0.5733 $\pm$ 0.0016
3. + class_weight=balanced	0.5741 $\pm$ 0.0024
4. + C tuned (C=0.3)	0.5751 $\pm$ 0.0028
5. + word(1,2) UNION char(3,5) [50k total]	0.5830 $\pm$ 0.0019
6. + CalibratedClassifierCV(LinearSVC)	0.5849 $\pm$ 0.0032

TABLE VI
TOP 5 OF 21 CANDIDATES TESTED, VIEW A (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
`ridge_calibrated`	linear	0.5652 $\pm$ 0.0064	0.6688	0.08
`logreg`	linear	0.5613 $\pm$ 0.0089	0.6599	1.40
`bagging_linsvc`	tree_ensemble	0.5601 $\pm$ 0.0013	0.6631	0.08
`linsvc_calibrated`	linear	0.5596 $\pm$ 0.0065	0.6642	0.08
`knn_k15`	instance_based	0.5541 $\pm$ 0.0069	0.6199	0.00

TABLE VII
TOP 5 OF 21 CANDIDATES TESTED, VIEW B (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
`ridge_calibrated`	linear	0.5542 $\pm$ 0.0076	0.6446	0.06
`logreg`	linear	0.5522 $\pm$ 0.0076	0.6381	0.58
`linsvc_calibrated`	linear	0.5519 $\pm$ 0.0074	0.6420	0.05
`linsvc_softmax`	linear	0.5488 $\pm$ 0.0065	0.6359	0.01
`bagging_linsvc`	tree_ensemble	0.5471 $\pm$ 0.0065	0.6377	0.05

TABLE VIII
LABEL-BUDGET SENSITIVITY: TEAM’S DEPLOYED MODEL VS. BEST-FOUND MODEL (MEAN OVER 3 SEEDS).

Labeled seed	Team A	Best A	Team B	Best B
300	0.5376	0.5466	0.5176	0.5126
600	0.5534	0.5652	0.5522	0.5542
900	0.5669	0.5813	0.5701	0.5729
1200	0.5908	0.6017	0.5830	0.5848

TABLE IX
SPELLING-NOISE ROBUSTNESS: MACRO-F1 UNDER INJECTED PHONETIC-VARIANT CORRUPTION (MEAN OVER 3 SEEDS).

Noise rate	Team A	Best A	Team B	Best B
0%	0.5534	0.5652	0.5522	0.5542
20%	0.5486	0.5614	0.5516	0.5535
35%	0.5444	0.5554	0.5524	0.5548

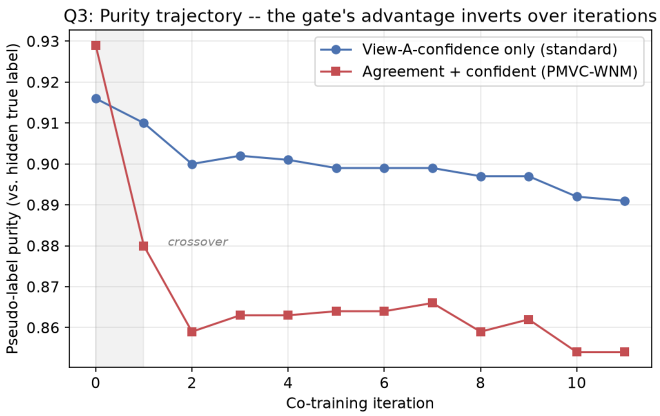

Fig. 1. Pseudo-label purity by co-training iteration, seed 42. The agreement gate's purity advantage appears only at iteration 0 and inverts by the final iteration.

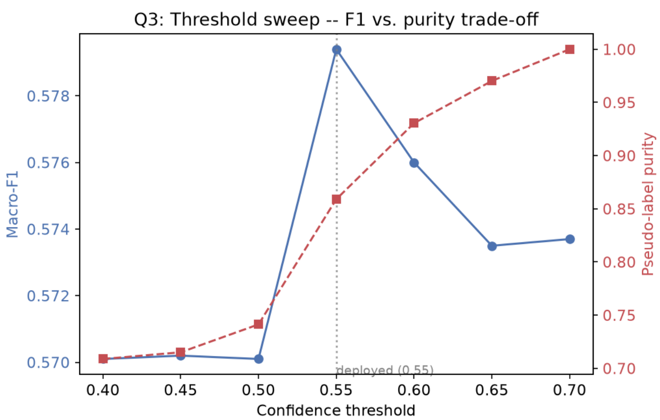

Fig. 2. Macro-F1 (left axis) and pseudo-label purity (right axis) vs. confidence threshold on the deployed agreement gate.

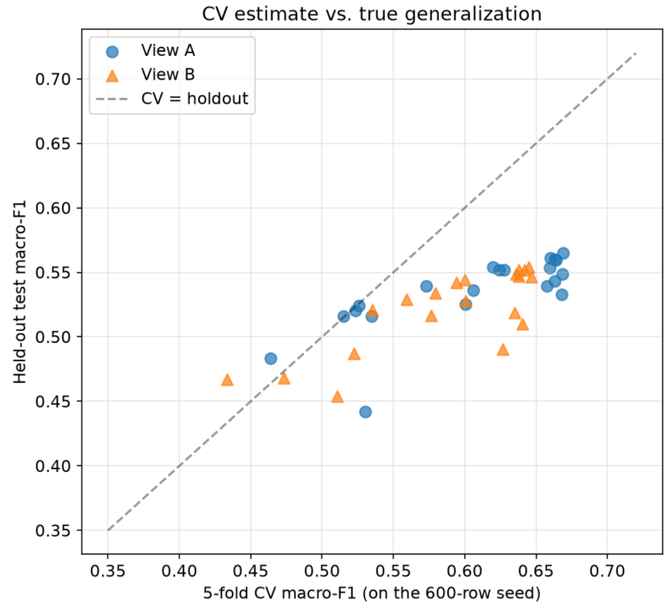

Fig. 3. Every candidate's 5-fold CV estimate vs. its held-out test score. Points below the diagonal generalize worse than their CV estimate suggested.

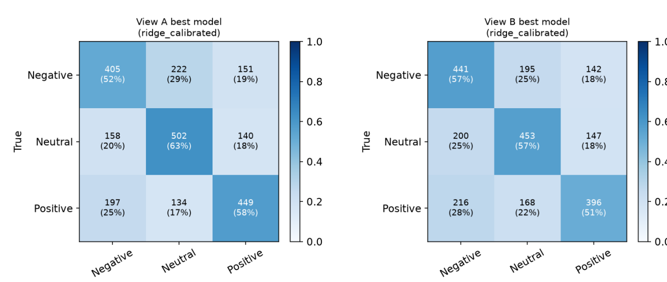

Fig. 4. Row-normalized confusion matrices, best model per view, seed 42.